SET A

CSE 341: Microprocessors
Department of Computer Science and Engineering
Brac University

Examination: Midterm Semester: Spring 2024 Duration: 1 Hour 20 Minutes Full Marks: 30

ANSWER ALL THE QUESTIONS

[CO1] Question 1 [6 marks]

1. **Explain** the significance of the instruction queue in the 8086 microprocessor. How does it contribute to improving performance? [2 marks]
2. **Describe** the purpose and operation of the segment registers in the 8086 architecture. How are they different from general-purpose registers? [3 marks]
3. Briefly **explain** in which scenario the instruction queue fails [1 mark]

[CO2] Question 2 [6 marks]

1. **Deduce** the equivalent machine code of the given instruction:
   a. MOV [BX + SI + 5312h], AX.
   Your answer should be in hexadecimal. [3]

2. **Explain** with reasoning which category of addressing modes the given instructions fall into. **MOV CL, [BX]**
   a. MOV CL, [BX+SI]
   b. RET [1234h]

[CO1] Question 3 [6 marks]

1. **Deduce** the maximum size of a memory device compatible with an 8086 microprocessor. Also, explain the consequences of utilizing an 8 GB RAM module with a 32-bit microprocessor. [2+1=3].
2. Assume you have a circuit where 8086 is currently the bus master. A DMA controller

device wants to initiate a DMA operation so it needs to acquire bus master control. Mention the relevant pin signals in the process of the DMA controller becoming the new bus master [2]. 3. **Explain** the functionality of the ALE pin in 8086 [1].

[CO1] Question 4 [6 marks]

1. **MOV AX, AFCFh**
   - **a. MOV BX, FFXYh**
   - **b. ADD AX, BX**

A. Determine the value of X and Y such that **both PF and AF are 1 and FFXY** will be **maximum after satisfying the given constraint**. [3 marks]

B. Now **calculate** the values of the OF, ZF, SF, and CF. You have to explain your answers with proper reasoning. [3 marks]

[CO1] Question 5 [6 marks]

Given DS = 2000h, CS = 3000h, SS = 1000h, SI = 1005h, IP = 1234h, SP = 2342h.

Address	Data
12345	12h
12344	23h
12343	34h
12342	45h
12341	56h

1. **Deduce** the instruction that loads data from the top of the stack into the AX register. [1 mark] 2. From the given table of the stack section **determine** what data will be stored in the AX register. [NB: For every byte of data loaded from the stack top, the value of SP increases by 1] [2 marks] 3. i. **Calculate** the physical address corresponding to the offset register SI. [2 marks] ii. Using your answer to part C i. find the **2nd highest and 3rd lowest logical addresses**. [1 mark]